

Windows Endpoint Threat Detection Report

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Executive Summary

This case study demonstrates a Windows endpoint threat hunting investigation performed using the Elastic Stack (ELK). Windows Security Event Logs were analyzed in Kibana to identify indicators of compromise including external device activity, suspicious NTLM authentication, and malicious service installation. Correlating multiple Event IDs enabled reconstruction of a likely attack sequence involving unauthorized authentication followed by persistence using an encoded PowerShell payload.

Objective

The objective was to identify evidence of endpoint compromise by examining Windows event logs, correlating related events, and documenting the attack lifecycle using industry-standard investigation techniques.

Environment

Platform: Elastic Stack (ELK)

Visualization: Kibana Discover

Data Source: Windows Security Event Logs

Investigation Method: Event ID correlation and manual log analysis.

Investigation Methodology

Three targeted investigations were performed.

1. External device detection using Event ID 6416.
2. Authentication analysis using Event ID 4624 while excluding Kerberos events using:
EventID:4624 AND NOT Message:"*Kerberos*"
3. Service installation analysis using Event ID 7045.

Each event was validated using hostnames, IP addresses, user accounts, authentication packages, workstation names, service names, and executable paths.

Evidence and Findings

Finding 1: Event ID 6416 identified a newly connected external USB device on IT04.YOLP.com (10.0.1.14).

Finding 2: Event ID 4624 analysis isolated successful NTLM (NtLmSsp) Logon Type 3 events. User 'droseberry' authenticated from workstation 5e9cQ0WHG4RFhNv.YOLP.com against IT03.YOLP.com, behavior consistent with Pass-the-Hash.

Finding 3: Event ID 7045 identified installation of a randomly named Windows service (flpWsoaatgwnVPnj). The service executed 'powershell.exe -nop -w hidden' with a Base64-encoded payload under the LocalSystem account, indicating malicious persistence.

Security Impact

Correlating Windows Event IDs 6416, 4624, and 7045 demonstrates how endpoint compromise can be reconstructed using security analytics. The workflow is representative of investigations performed by Security Operations Centers (SOCs) to identify compromised hosts and attacker persistence.

Skills Demonstrated

Windows Event Log Analysis

Elastic Stack (ELK)

Kibana Discover

Threat Hunting

Endpoint Detection

Pass-the-Hash Detection

PowerShell Analysis

Incident Response

Security Analytics

Conclusion

This investigation successfully identified multiple indicators of compromise by correlating Windows Event Logs. The evidence supports a probable attack sequence involving unauthorized NTLM authentication followed by PowerShell-based persistence, illustrating a practical enterprise threat hunting workflow.